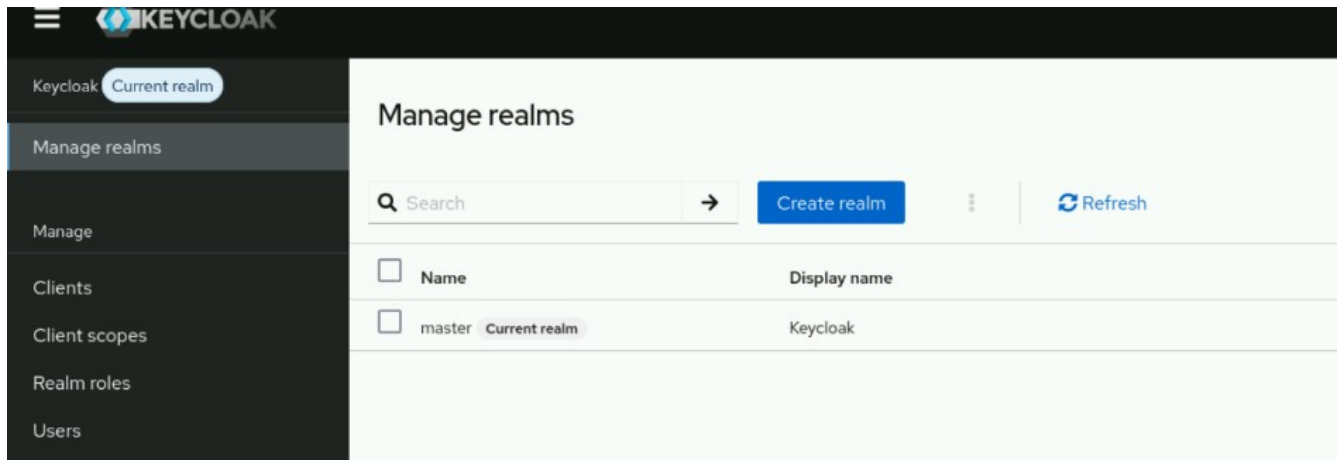


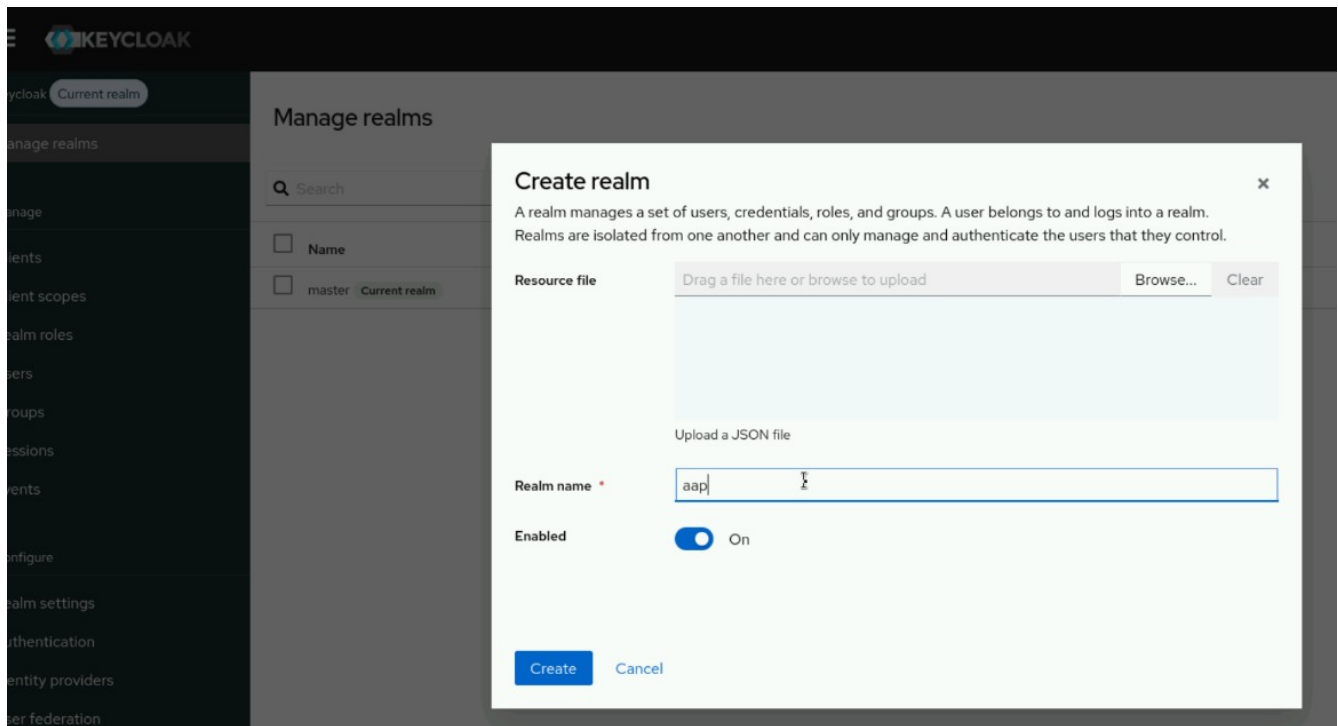
Setup Keycloak on aap server

1. [student@workstation ~]\$ ssh [devops@aap](#)
2. mkdir oidc ; cd oidc
3. podman cp automation-gateway:/etc/ansible-automation-platform/gateway/gateway.key keycloak.key
4. podman cp automation-gateway:/etc/ansible-automation-platform/gateway/gateway.crt keycloak.crt
5. chmod 644 keycloak.*
6. sudo dnf install pip -y
7. pip3 install podman-compose
8. podman compose -f podman-compose.yml up -d
9. sudo firewall-cmd --add-port=8443/tcp --permanent
10. sudo firewall-cmd --reload

Workstation Firefox Browser

1. <https://aap.lab.example.com:8443>
2. Login as USERNAME – admin // PASSWORD – admin
3. Click “Create Realm”





Create Client:

aap	Current realm
Manage realms	
Manage	
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Client scopes	
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Configure	
Realm settings	

Clients

Clients are applications and services that can request authentication of a user. [Learn more](#)

Clients list

Initial access token

Client registration

Search for client

Create client

Import client

Refresh

Client ID	Name	Type	Description	Home URL
account	Account	OpenID Connect	—	https://aap.lab.example.com:8443/auth/realms/aap/account/
account-console	Account Console	OpenID Connect	—	https://aap.lab.example.com:8443/auth/realms/aap/account/
admin-cli	Admin CLI	OpenID Connect	—	—
broker	Broker	OpenID Connect	—	—
realm-management	Realm Management	OpenID Connect	—	—
security-admin-console	Security Admin Console	OpenID Connect	—	https://aap.lab.example.com:8443/auth/admin/aap/console/

aapCurrent realm

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Create client

Clients are applications and services that can request authentication of a user.

1 General settings

Client type ⓘ

OpenID Connect

Client ID ⓘ

aap

Name ⓘ

Description ⓘ

Always display in UI ⓘ

☒ On

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Cancel

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Clients are applications and services that can request authentication of a user.

1 General settings

2 Capability config

Client authentication ⓘ

☒ On

Authorization ⓘ

☐ Off

Authentication flow

☒ Standard flow ⓘ

☐ Implicit flow ⓘ

☐ Standard Token Exchange ⓘ

☐ JWT Authorization Grant ⓘ

☐ OAuth 2.0 Device Authorization Grant ⓘ

☐ OIDC CIBA Grant ⓘ

☒ Direct access grants ⓘ

☐ Service account roles ⓘ

Require PKCE ⓘ

☐ Off

Require DPoP bound tokens ⓘ

☐ Off

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Clients are applications and services that can request authentication of a user.

1 General settings

2 Capability config

3 Login settings

Root URL ⓘ

https://aap.lab.example.com/

Home URL ⓘ

Valid redirect URIs ⓘ

https://aap.lab.example.com/*

+ Add valid redirect URIs

Valid post logout redirect URIs ⓘ

+ Add valid post logout redirect URIs

Web origins ⓘ

https://aap.lab.example.com/*

+ Add web origins

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Clients > Client details

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OpenID Connect

Enabled ⓘ

Action ⌵

SettingsKeysCredentialsRolesClient scopesSessionsAdvancedEvents

SetupEvaluate

⌵ Name

Q Search by name

→

Add client scope

Change type to ⌵

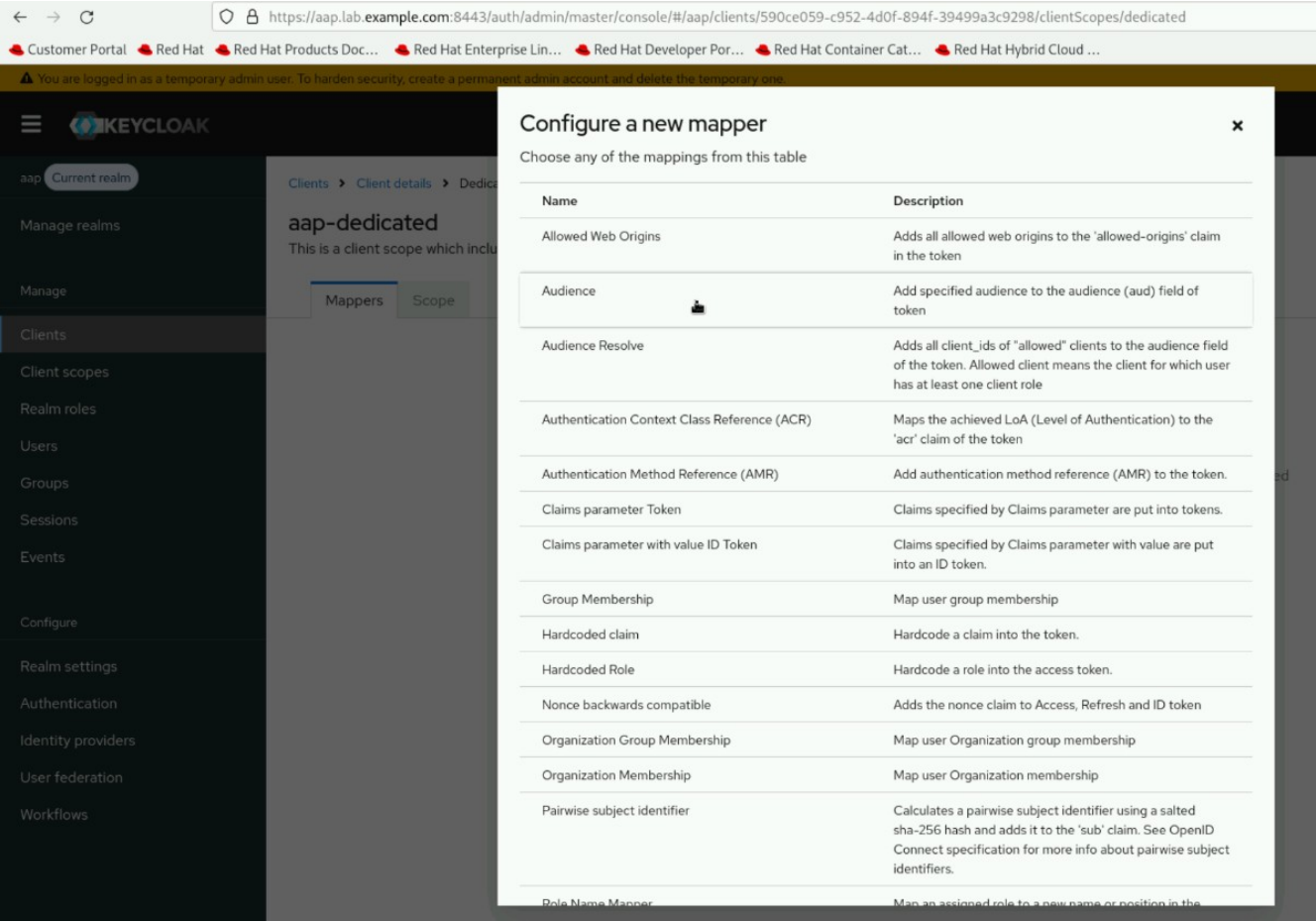
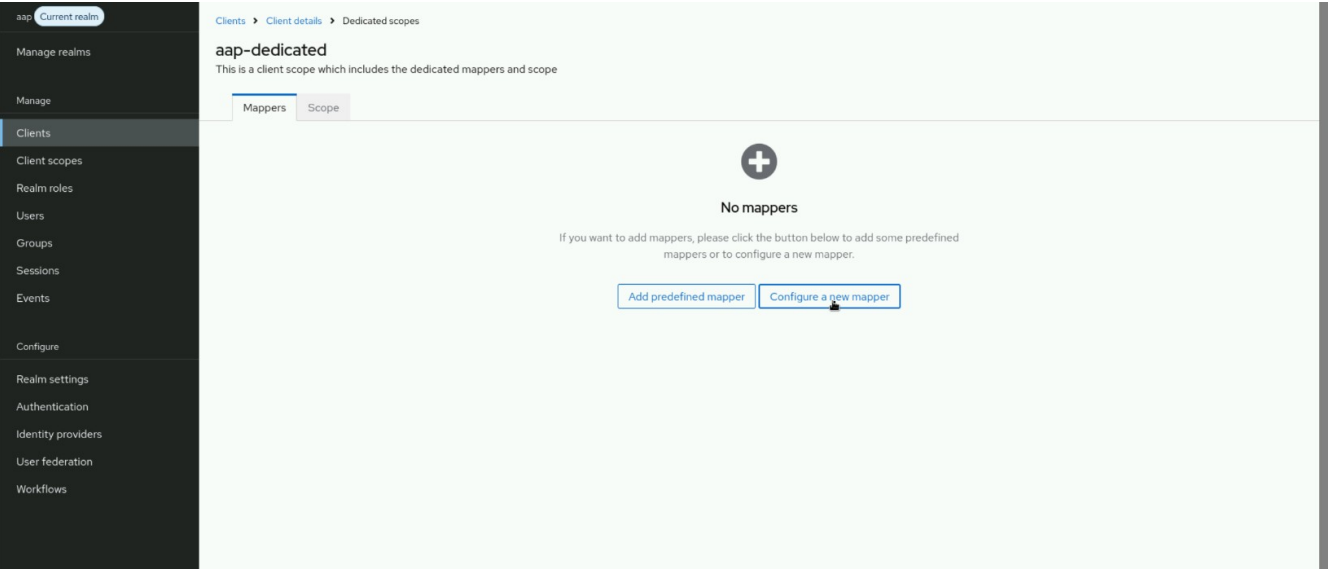
:

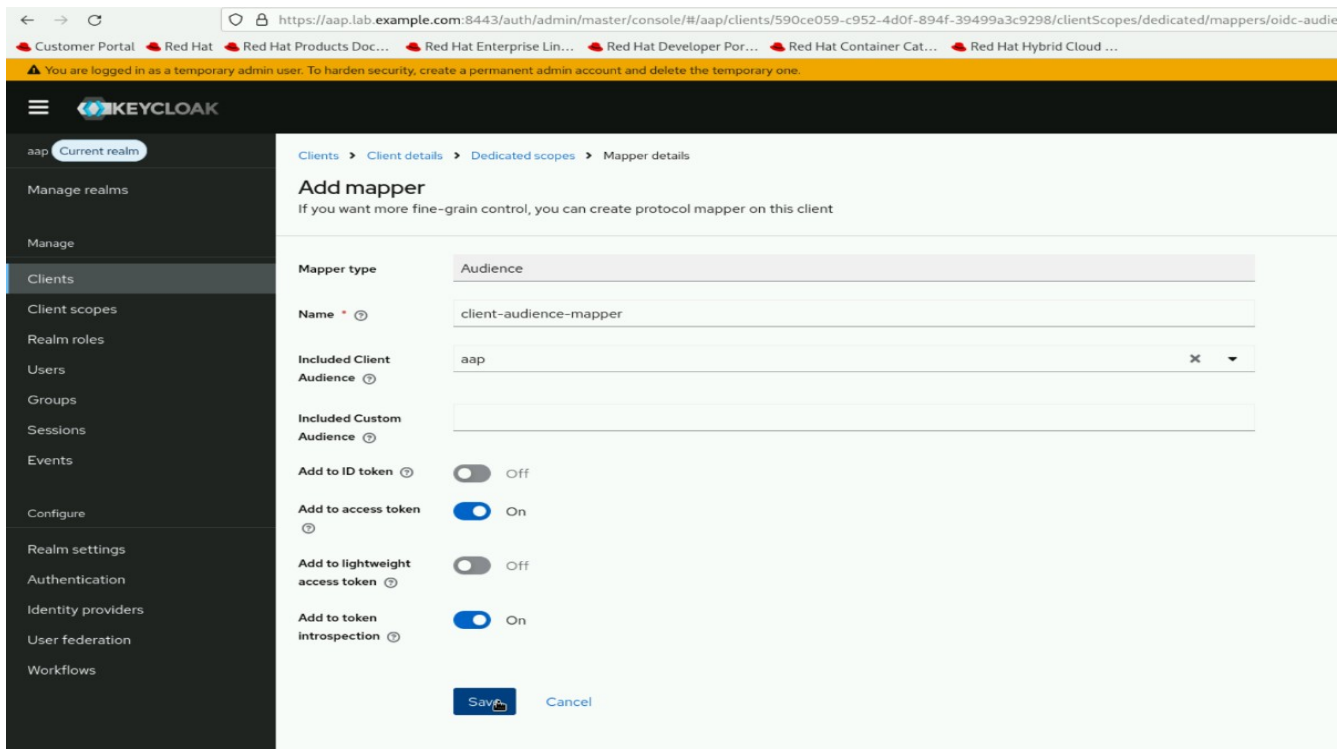
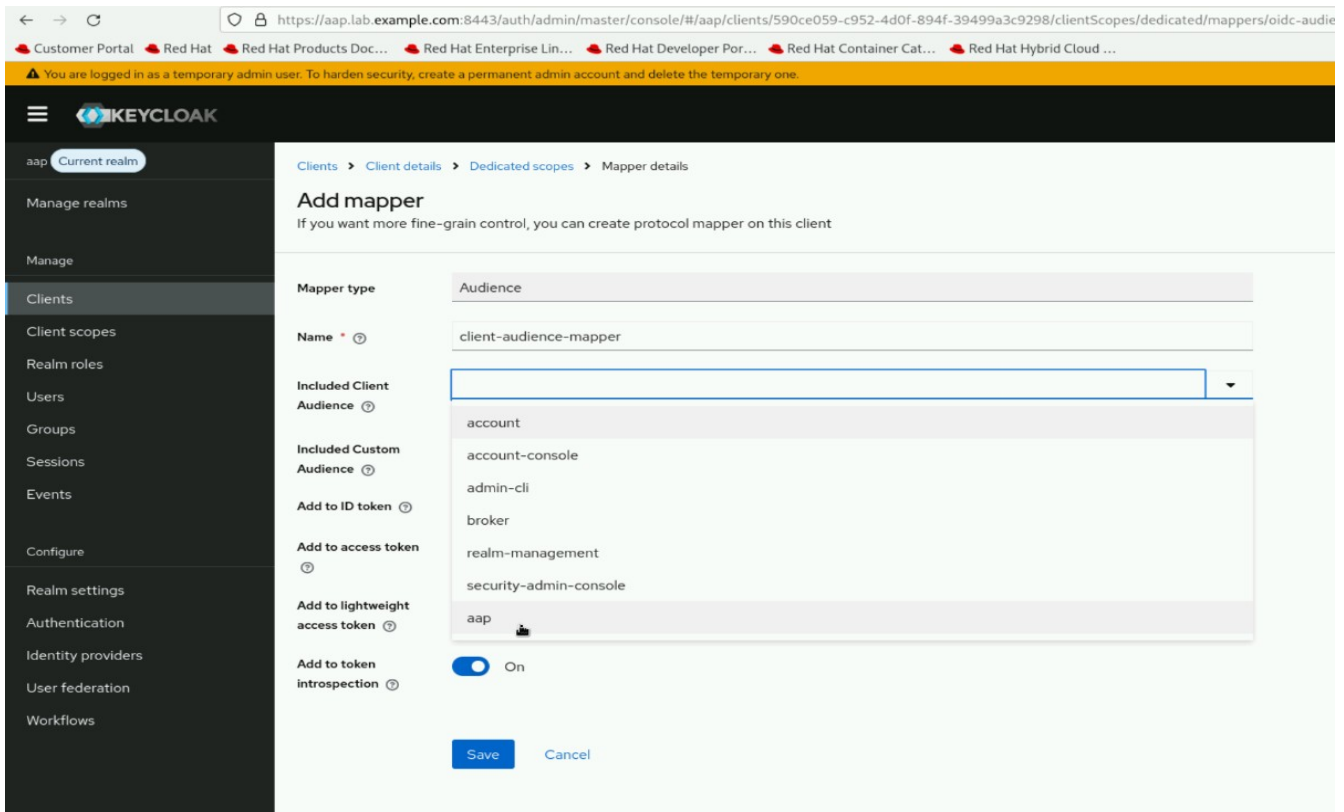
Refresh

1-10 ⌵ ⌵ ⌵

Assigned client scope	Assigned type	Description
☐ aap-dedicated	None ⌵	Dedicated scope and mappers for this client
☐ acr	Default ⌵	OpenID Connect scope for add acr (authentication context class reference) to the token
☐ address	Optional ⌵	OpenID Connect built-in scope: address
☐ basic	Default ⌵	OpenID Connect scope for add all basic claims to the token
☐ email	Default ⌵	OpenID Connect built-in scope: email
☐ microprofile-jwt	Optional ⌵	Microprofile - JWT built-in scope
☐ offline_access	Optional ⌵	OpenID Connect built-in scope: offline_access
☐ organization	Optional ⌵	Additional claims about the organization a subject belongs to
☐ phone	Optional ⌵	OpenID Connect built-in scope: phone

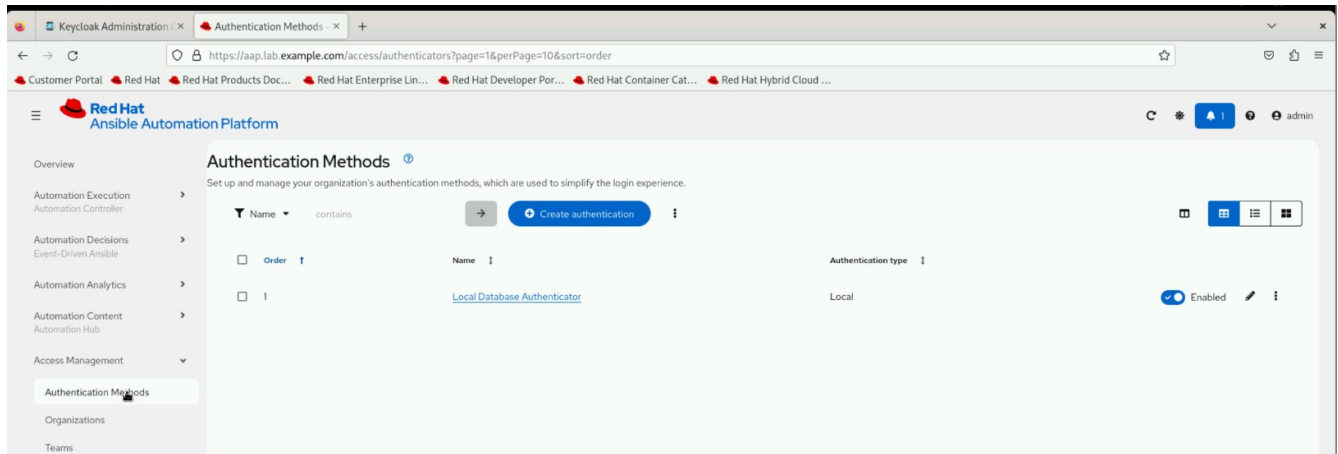
https://aap.lab.example.com:8443/auth/admin/master/console/#/aap/clients/590ce059-c952-4d0f-894f-39499a3c9298/clientScopes/dedicated





Login in to aap.lab.example.com

1. Open Another Browser Tab on firfox and <https://aap.lab.example.com> with admin / redhat
- 2.



3. Click on “Create Authentication”

Keycloak Access Token URL

<https://aap.lab.example.com:8443/auth/realms/aap/protocol/openid-connect/token>

Keycloak Provider URL

<https://aap.lab.example.com:8443/auth/realms/aap/protocol/openid-connect/auth>

Keycloak Administration

Create authentication

https://aap.lab.example.com/access/authenticators/create

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Red Hat Ansible Automation Platform

admin

Overview

Automation ExecutionAutomation Controller

Automation DecisionsEvent-Driven Ansible

Automation Analytics

Automation ContentAutomation Hub

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Users

Roles

OAuth Applications

Authentication Methods

Create authentication

Create authentication

Name *

Keycloak

Auto migrate users from

Select authenticators

Authentication type *

Keycloak

Authentication details

Keycloak Access Token URL

https://aap.lab.example.com:8443/auth/realms/aap/protocol/openi ...

Keycloak Provider URL

https://aap.lab.example.com:8443/auth/realms/aap/protocol/openi ...

Keycloak OIDC Key *

aap

Keycloak Public Key *

Enter Keycloak Public Key

Keycloak OIDC Secret *

Enter Keycloak OIDC Secret

Groups Claim

Group

Additional Authenticator Fields

YAMLJSON

Keycloak OIDC Key
The OIDC key (Client ID) from your Keycloak installation.

Keycloak Administration

Create authentication

https://aap.lab.example.com:8443/auth/admin/master/console/#/aap/realms/settings/keys

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You are logged in as a temporary admin user. To harden security, create a permanent admin account and delete the temporary one.

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Realm settings are settings that control the options for users, applications, roles, and groups in the current realm. [Learn more](#)

GeneralLoginEmailThemesKeysEventsLocalizationSecurity defensesSessionsTokensClient policiesUser profileUser registration

Keys listProviders

Active keys

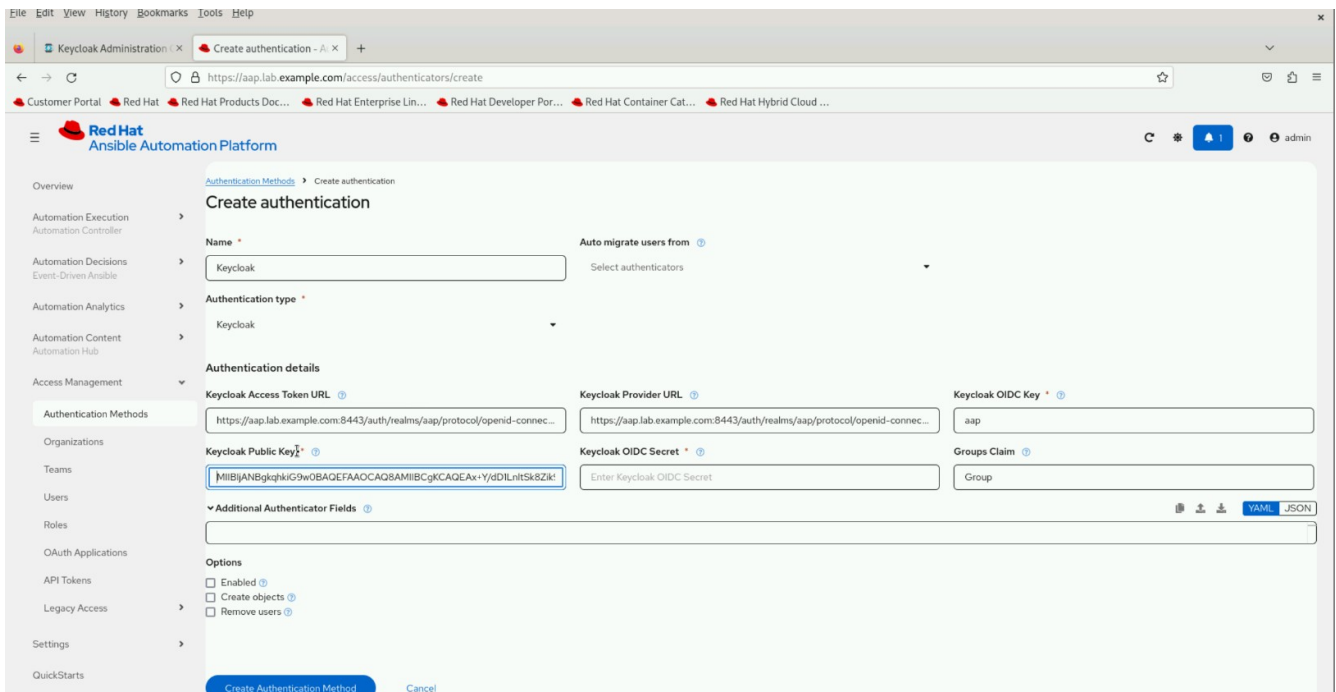
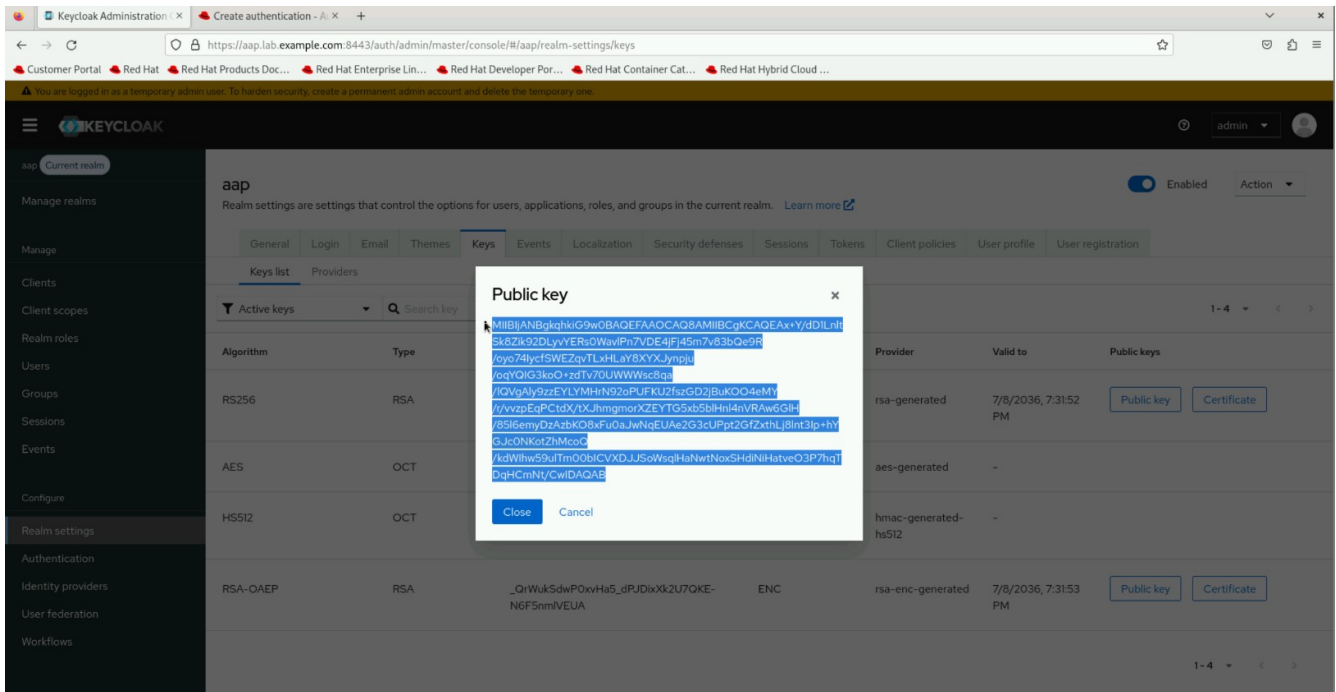
Search key

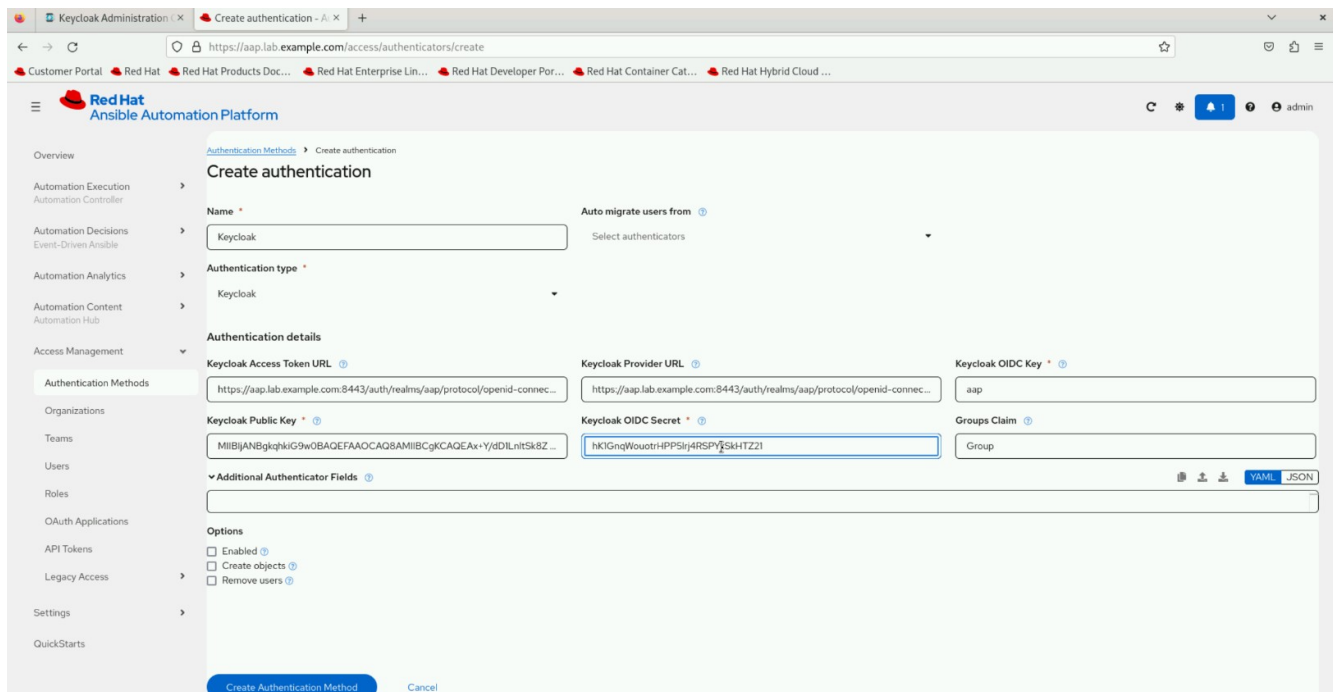
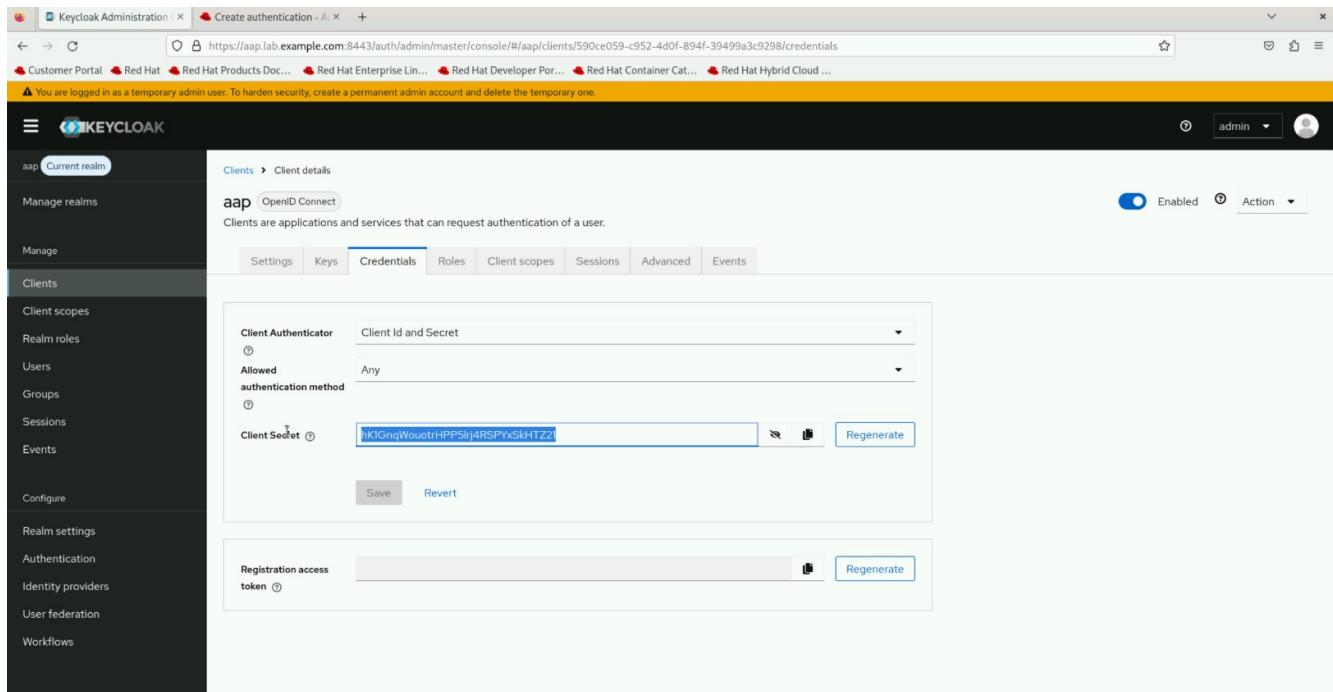
Refresh

1-4

Algorithm	Type	Kid	Use	Provider	Valid to	Public keys
RS256	RSA	h1J-Qxrrs-GAgvEHw9jSIS6ZFXymOJCcqVIF2RDc0mw	SIG	rsa-generated	7/8/2036, 7:31:52 PM	Public key Certificate
AES	OCT	bbede63a-6449-4ed2-af7d-e6442d1621a5	ENC	aes-generated	-	
HS512	OCT	609ef817-0386-490f-9020-accdd09eb48b	SIG	hmac-generated-hs512	-	
RSA-OAEP	RSA	_OrWukSdwP0xvHa5_dPJDiXk2U7QKE-N6F5nmIVEUA	ENC	rsa-enc-generated	7/8/2036, 7:31:53 PM	Public key Certificate

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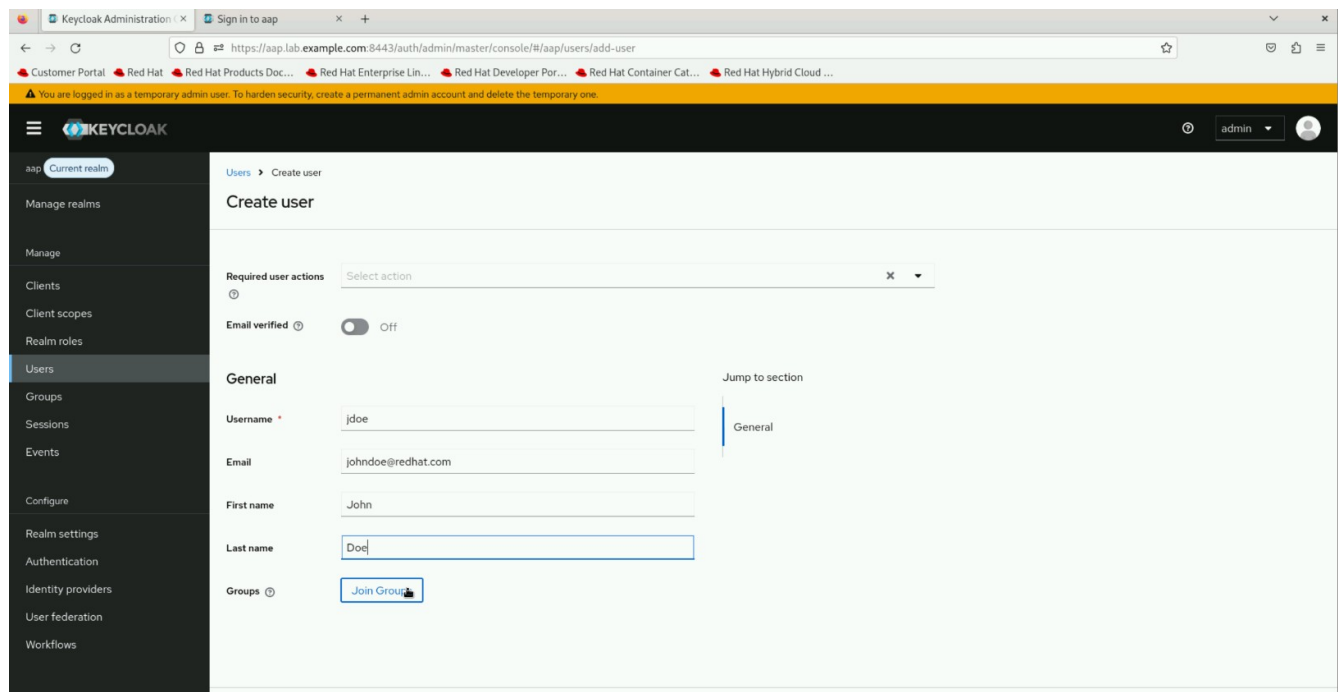
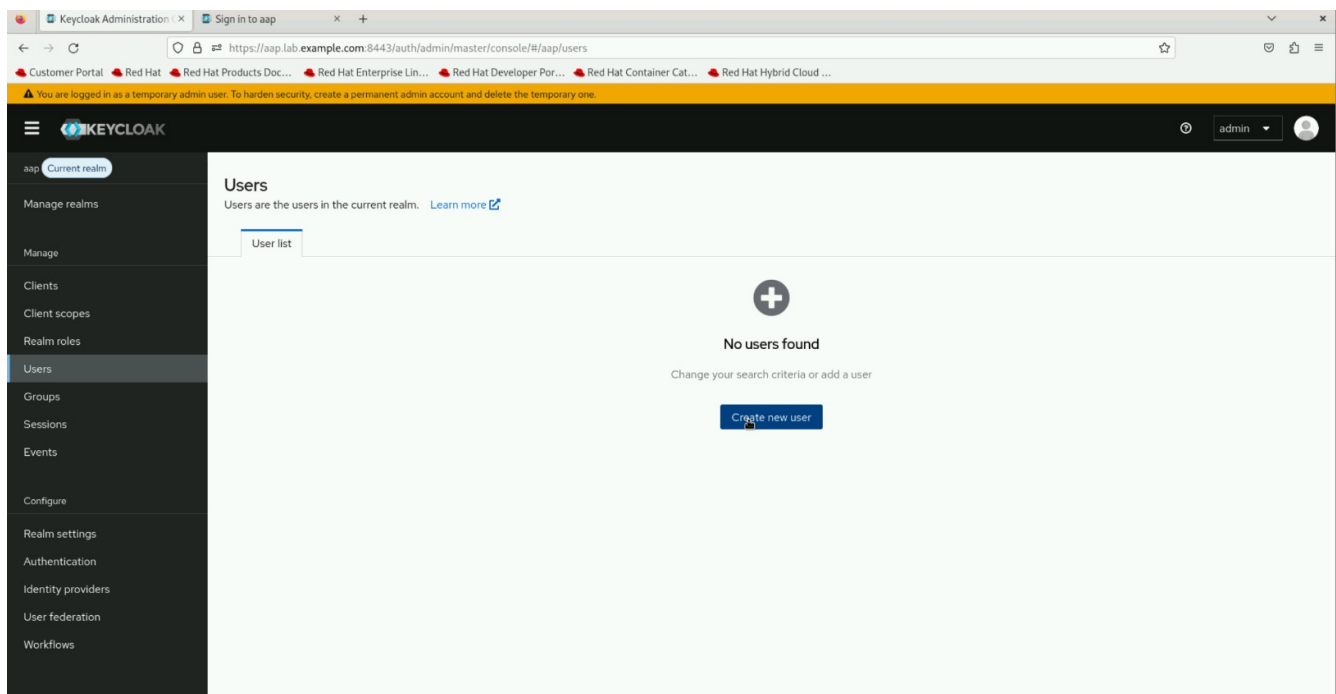
The screenshot displays the Red Hat Ansible Automation Platform Keycloak Administration console. The browser address bar shows the URL `https://aap.lab.example.com/access/authenticators/create`. The left sidebar contains a navigation menu with the following items: Overview, Automation Execution (Automation Controller), Automation Decisions (Event-Driven Ansible), Automation Analytics, Automation Content (Automation Hub), Access Management (selected), Authentication Methods (selected), Organizations, Teams, Users, Roles, OAuth Applications, API Tokens, Legacy Access, Settings, and QuickStarts. The main content area is titled 'Create authentication' and includes the following sections:

- Authentication Methods**: A breadcrumb trail showing 'Authentication Methods' > 'Create authentication'.
- Name**: A text input field containing 'Keycloak'.
- Auto migrate users from**: A dropdown menu with the option 'Select authenticators'.
- Authentication type**: A dropdown menu with the option 'Keycloak'.
- Authentication details**:
 - Keycloak Access Token URL**: A text input field containing `https://aap.lab.example.com:8443/auth/realms/aap/protocol/openid-connect...`.
 - Keycloak Provider URL**: A text input field containing `https://aap.lab.example.com:8443/auth/realms/aap/protocol/openid-connect...`.
 - Keycloak OIDC Key**: A text input field containing 'aap'.
 - Keycloak Public Key**: A text input field containing `MIIjANBgglqhkiG9w0BAQFAAQASAMIBGKCAQEAx+YdDLNtSk8Z...`.
 - Keycloak OIDC Secret**: A text input field containing `hKIGnqWouotrHPP5Ijr4RSPriXskHTZl`.
 - Groups Claim**: A text input field containing 'Group'.
- Additional Authenticator Fields**: A section with a dropdown arrow and a text input field.
- Options**: A list of checkboxes:
 - ☒ Enabled
 - ☒ Create objects
 - ☒ Remove users

At the bottom of the page, there are two buttons: 'Create Authentication Method' and 'Cancel'.

The screenshot shows the Red Hat Ansible Automation Platform interface. The main content area displays the 'Keycloak' configuration details. The 'Authentication Methods' sidebar on the left lists various authentication methods, with 'Keycloak' selected. The 'Details' tab for Keycloak shows the following information:

Name	Type	Keycloak Access Token URL	Keycloak Provider URL
Keycloak	Keycloak	https://aap.lab.example.com:8443/auth/realms/aap/protocol/openid-connect/token	https://aap.lab.example.com:8443/auth/realms/aap/protocol/openid-connect/auth
Groups Claim	Keycloak OIDC Key	Keycloak Public Key	Keycloak OIDC Secret
Group	aap	MIIBjANBgkqhkiG9w0BAQEFAOCAQ8AMIIBgKCAQEAAx+YjdDlNtSk8Zk9ZDlyYERsOWavPh7VDE4fjH5m7v83bQe9R/oyo74lycfSWEZqVTLxHLay8XYKJympju/jeqQIK3keO+zd7v70UWWwsc8pa/OVG9AlYszzEYLYMHrN92oPUFRU2hzGD2BukOO4eM/Yt/vvzqPCtDx/TXJhmgmorXZEYTG5xb5bHn4nVRaw6GH/85i6emyDzAz8K08xFu0aJwNqEUAe2G3cUPpt2GFZxtHJ8lnt3lp+HYG-Jc0NKotZhmccO/kdWlhW59uTm00bCVXDJJSoWsqHhNwtNaxSHdNIHavteO3P7h7qDqHcmNt/CwIDAQAB	EncryptedS



Keycloak Administration

Sign in to aap

https://aap.lab.example.com:8443/auth/admin/master/console/#/aap/users/38dfc6e0-86c1-46b9-8a02-a3841b6d6b37/credentials

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No credentials

This user does not have any credentials. You can set password for this user.

Set password

Credential Reset

